

Multiple Choice (10 marks)

1. Use partial fraction expansion to find the inverse z-transform of $F(z) = \frac{2z^2 - 2z}{(z-3)(z-5)^2}$.
 - (a) $f[n] = 4n u[n] + [(5/6)n - 1] 6^n u[n]$
 - (b) $f[n] = 6n u[n] + [(2/3)n - 1] 3^n u[n]$
 - (c) $f[n] = 3^{(n+1)} u[n] - 5^n u[n]$
 - (d) $f[n] = 3n u[n] + [(4/5)n - 1] 5^n u[n]$
 - (e) $f[n] = [(2/3)n + 1] (3/2)^n u[n] - 2^{(n-1)} u[n]$
2. A D-flip-flop is said to be transparent when
 - (a) the output is independent of the clock
 - (b) the output is constantly changing
 - (c) the output is triggered by a rising edge clock
 - (d) the output is LOW
 - (e) the output follow input
3. For a short dipole antenna, evaluate the maximum effective aperture A_{em} .
 - (a) 0.119 wavelength
 - (b) 0.130 wavelength
 - (c) $0.112 \text{ wavelength}^2$
 - (d) 0.100 wavelength
 - (e) $0.123 \text{ wavelength}^2$
4. A liquid-propellant rocket uses 22 lb of fuel per second and 300 lb of oxidant per second. The exhaust gases leave the rocket at 2,000 fps. Calculate the rocket thrust.
 - (a) 45,000 lb
 - (b) 18,000 lb
 - (c) 28,000 lb
 - (d) 30,000 lb
 - (e) 15,000 lb
5. The current through an inductor with inductance $L = 10^{-3} \text{ henry}$ is given as $i_L(t) = 0.1 \sin 10^6 t$. Find the voltage $v_L(t)$ across this inductor.
 - (a) $1000 \sin 10^6 t$

- (b) $100 \sin 10^6 t$
- (c) $50 \cos 10^6 t$
- (d) $1000 \cos 10^6 t$
- (e) $0.1 \cos 10^6 t$

True / False (10 marks)

Answer True or False

6. True or False: The optimal length of a half-wave dipole antenna for 300 MHz with a velocity factor of 1.0 is 0.6 meters.
7. True or False: The Q-factor of an antenna with a bandwidth of 0.8 MHz and a cut-off frequency of 44 MHz is 55.
8. True or False: The change in stored energy of a parallel plate capacitor after inserting a dielectric slab is typically 0.99 times the original energy.
9. True or False: The probability of rolling either a 2 or a 5 on a single die is $1/2$.
10. True or False: The potentiometer method of DC voltage measurement does not load the circuit, allowing for more accurate voltage readings.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of burning hydrogen with air in an adiabatic burner, and calculate the percentage of theoretical air required for complete combustion at a flame temperature of 2000 K, given that both hydrogen and air are supplied at 25°C.
11. Discuss the relationship between an antenna's directivity and its maximum effective aperture, given a directivity of 50 and a wavelength of 4 meters. Calculate the effective aperture and explain its significance in antenna design.

End of Paper